

Dr. VISHNU BHUVANACHANDRAN

PhD | MSc — First Class Honours with Gold Medal | Associate Professor | Research Lead
Mechanical Engineering | AI / Machine Learning | Data Analytics | Sustainable Energy
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PERSONAL DETAILS

Nationality:	Indian
Current Location:	Dundalk, Co. Louth, Ireland (EU-based)
Languages:	English (Fluent), Malayalam (Native), Tamil (Proficient)
Availability:	Available immediately; open to relocation

RESEARCH INTERESTS

Sustainable Energy Systems • EV Battery Thermal Management • Environmental Safety & Occupational Health • AI/ML Applications in Mechanical Engineering • Computational Data Analysis • Emission Reduction & Green Technologies • Socio-economic Data Modelling

ACADEMIC PROFILE

An academically accomplished Mechanical Engineer with a PhD from the National Institute of Technology Tiruchirappalli (NIT-T) one of India's premier research institutions and an MSc in Data Analytics awarded with First Class Honours and Gold Medal from Dundalk Institute of Technology, Ireland. Holder of 5 Indian patents and author of 15+ peer-reviewed publications. Former Associate Professor with 5 years of undergraduate and postgraduate teaching experience, and institutional leadership roles including Assistant Nodal Officer of IEDC and Convener of the Institute Innovation Council. Research expertise spans sustainable energy, EV thermal systems, environmental safety, and emission control, complemented by advanced skills in AI/ML, Python, and Power BI. Recipient of multiple competitive research fellowships from the Indian Government's Department of Atomic Energy (DAE). International speaker at two IIW (International Institute of Welding) Assemblies.

EDUCATION

Dundalk Institute of Technology

Ireland

MSc in Data Analytics — First Class Honours with Gold Medal

2024 – 2025

- Modules: Machine Learning, GDPR & Data Protection, Data Visualisation, Statistical Analysis, Big Data Analytics
- Projects: ML-based disease prediction (2.3M data points), cyber threat dashboards, transformer-based time series analysis
- European academic qualification; highest distinction awarded

National Institute of Technology Tiruchirappalli (NIT-T)

India

PhD in Mechanical Engineering — Environmental Safety & Occupational Health

2014 – 2019

- Research: Cr(VI) emission reduction achieved 70% reduction through novel experimental and analytical methodologies
- Funded by Department of Atomic Energy, Government of India (IGCAR fellowship)
- 10+ peer-reviewed international publications; 5 Indian patents

University of Kerala

India

M.Tech in Thermal Science

2011 – 2013

University of Kerala

India

B.Tech in Mechanical Engineering

2005 – 2009

PROFESSIONAL EXPERIENCE

Starilum Limited

Data Analyst

Drogheda, Ireland

June 2025 – Present

- Maintain and manage complex datasets using SQL and Excel, ensuring data accuracy through daily validation aligned with internal governance standards.
- Build and maintain Power BI dashboards integrating multiple data sources, delivering actionable insights for cross-functional reporting.
- Generate regular and ad-hoc reports with documented data transformation steps to support structured, accountable decision-making.
- Achieved an 80% increase in organisational research capabilities within 5 months by enhancing reporting infrastructure and integrating disparate data sources.

Marian Engineering College (APJ Abdul Kalam Technological University)

Kerala, India

Associate Professor & Research Lead — Mechanical Engineering

Sept 2019 – Aug 2024

- Delivered undergraduate and postgraduate lectures in Mechanical Engineering, Thermal Sciences, and Engineering Data Analysis; supervised final-year projects and research dissertations.
- Served as Assistant Nodal Officer of the Innovation and Entrepreneurship Development Centre (IEDC) and Convener of the Institute Innovation Council (IIC), driving an institutional culture of innovation and student entrepreneurship.
- Founded the Energy Research Lab and led two government-funded projects: INR 12 Lakhs EV Battery Cooling System research and INR 28 Lakhs Handloom & Coir Cluster Mapping project.
- Coordinated a 30+ member interdisciplinary team on the Electric Go-Kart project, achieving national-level recognition and a 90% improvement in innovation metrics.
- Published 10+ peer-reviewed papers, secured 5 Indian patents, and represented the institution at national and international academic forums.

Indira Gandhi Centre for Atomic Research (IGCAR) — DAE, Govt. of India

India

Senior Research Fellow

May 2014 – Jan 2019

- Conducted doctoral research under a DAE Government of India fellowship within one of India's premier nuclear research institutions.
- Designed experimental protocols for emission characterisation and control, achieving a 70% reduction in Cr(VI) emissions through rigorous data-driven analysis.
- Produced compliance and performance reports for international safety oversight panels (NIOSH, AWS standards); presented findings to multidisciplinary scientific committees.
- Collaborated across engineering, chemistry, and safety departments, achieving a 90% improvement in process efficiency on sustainability-focused projects.

PUBLICATIONS & INTELLECTUAL PROPERTY

Peer-Reviewed Journal Publications (10+)

- 15+ peer-reviewed publications in indexed international journals (SCI/Scopus) in mechanical engineering, environmental safety, emission control, and sustainable materials. Full list available on request.

Patents

- 5 granted Indian Patents in sustainable engineering processes, emission control technologies, and materials innovation. Patent details available on request.

International Conference Presentations

- Speaker at two IIW (International Institute of Welding) General Assemblies – an internationally recognised forum bringing together welding science and engineering professionals from over 50 countries.

FUNDED RESEARCH PROJECTS

EV Battery Cooling System — Research Grant

2019 – 2024

Principal Investigator | INR 12 Lakhs

Marian Engineering College

- Designed and led experimental research on thermal management systems for EV batteries, directly relevant to Gulf region sustainability and EV adoption goals under Vision 2030.

Handloom & Coir Cluster Mapping — Government-Funded Project

2019 – 2024

Project Lead | INR 28 Lakhs

Marian Engineering College

- Integrated socio-economic and geospatial datasets to produce policy-relevant analysis and reports for regional government planning bodies.
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TECHNICAL & AI / DATA ANALYTICS SKILLS

AI & Machine Learning: Supervised/Unsupervised Learning, Time Series Models, Imbalanced Data Handling, Predictive Modelling

Programming: Python (Pandas, NumPy, Scikit-learn, Matplotlib), R, SQL, HTML, JavaScript

Data Visualisation: Power BI, Tableau, Shiny (R), Seaborn

Databases: SQL, PostgreSQL, Neo4j (Graph Databases)

Engineering Tools: MATLAB, Minitab, Excel (Advanced)

Cloud & Platforms: AWS (Fundamentals), Google Workspace

Research Methods: Statistical Analysis, Experimental Design, Regression Analysis, Trend Forecasting

Other: Agile Methodology, GDPR Principles, Ethical Research Documentation, Technical Writing

KEY ACHIEVEMENTS & AWARDS

- Gold Medal — MSc Data Analytics, First Class Honours, Dundalk Institute of Technology, Ireland (2025): Highest academic distinction awarded by a European institution.
 - 5 Granted Indian Patents: Novel contributions in sustainable engineering and emission control technologies.
 - 10+ Peer-Reviewed International Publications: Indexed journal articles (SCI/Scopus) establishing independent research credentials.
 - International Speaker — IIW General Assemblies: Presented research at two assemblies of the International Institute of Welding, a body with membership spanning 50+ countries.
 - DAE-IGCAR Research Fellowships: Multiple competitive fellowships from the Government of India's Department of Atomic Energy — a nationally prestigious distinction.
 - National Award — Electric Go-Kart Project: Team recognised at national level under research and innovation coordination.
 - 70% Cr(VI) Emission Reduction: Quantified, high-impact research outcome in environmental health and industrial safety.
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TEACHING PHILOSOPHY

I believe in an inquiry-led, data-informed approach to engineering education encouraging students to question assumptions, engage with real-world datasets, and develop both technical rigour and critical thinking. My interdisciplinary background across mechanical engineering, environmental science, and data analytics equips me to teach core engineering subjects while integrating contemporary AI and data tools into the curriculum, preparing students for the demands of the modern engineering landscape.

REFERENCES

Academic and professional references available on request.